

kafka-isotope

End-to-end **record tracing for Apache Kafka**, built entirely on public extension points — a **ProducerInterceptor** plus record headers. No broker changes, no agent, no vendor lock-in. Stamp a UUIDv7 trace id and origin metadata on first produce, accumulate a hop on every re-produce, and reconstruct latency / topology / hop-distribution from the headers (or from the optional Prometheus meters).

1.0 Modules

Module	Coordinate	Pulls	Use it for
isotope-core	<code>ai.signalroom:isotope-core</code>	Jackson, SLF4J (<code>kafka-clients</code> is <code>compileOnly</code>)	Trace propagation — interceptor, headers, consume markers.
isotope-metrics	<code>ai.signalroom:isotope-metrics</code>	isotope-core + Micrometer/Prometheus	Optional <code>/metrics</code> exporter for the stateless reports.

`isotope-core` never depends on a metrics library: emission routes through a no-op `IsotopeMetricsSink` until `isotope-metrics` registers the Prometheus one, so propagation runs with zero metrics overhead.

2.0 Install using Gradle

```
repositories {
    maven { url 'https://maven.pkg.github.com/j3-signalroom/kafka-isotope' }
}
dependencies {
    implementation 'ai.signalroom:isotope-core:0.17.1'
    implementation 'ai.signalroom:isotope-metrics:0.17.1' // optional — only for Prometheus
}
```

See [isotope-core/README.md](#) for the full adopter quickstart (registering the interceptor, adopting/marking on consume, starting the exporter).

Demo

A runnable reference pipeline (Confluent Platform / Confluent Cloud on Minikube, the seven Flink reports, and a one-command Prometheus + Grafana showcase) lives in the companion repo: [j3-signalroom/confluent-kafka-isotope](#).